

AviLight Ecological Impact and Diagnostics Report

Executive Summary

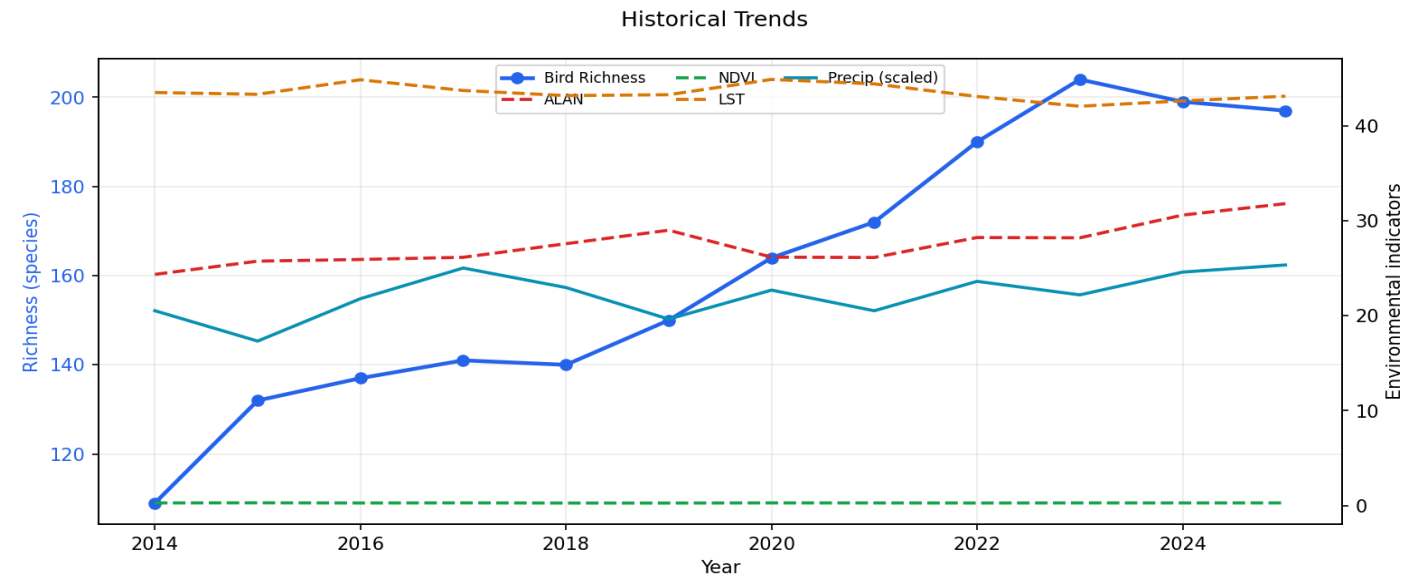
Run Context

Field	Value
Area	All Areas
Trend Years	2014-2025
Snapshot	2025-01
Generated	2026-04-19 17:00:42 UTC

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Ecological Impact View

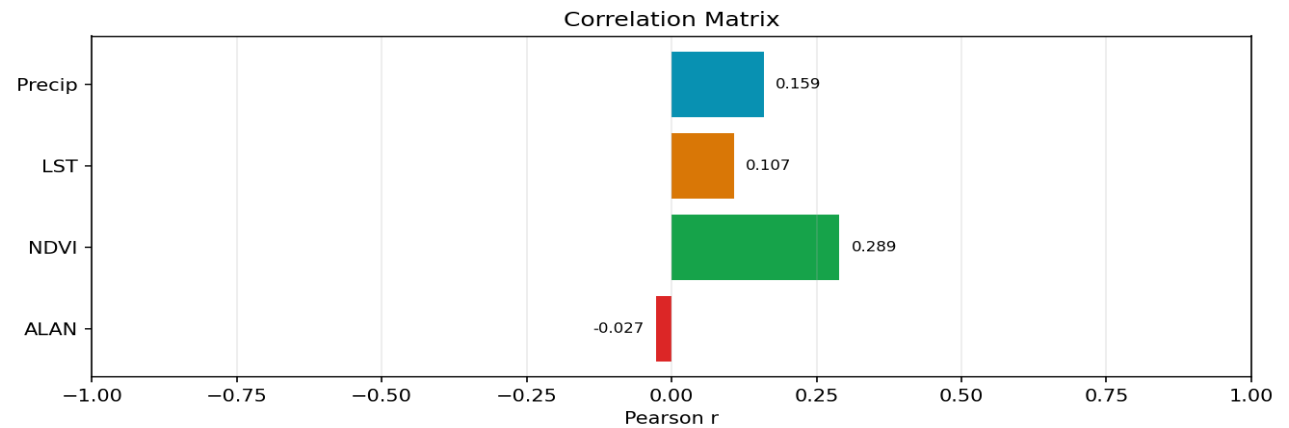
Chart: Long-Term Ecological Impact Trends



Numerical Values: Historical Trends

Year	Richness	ALAN	NDVI	LST	Precip
2014	109.00	24.35	0.2770	43.51	2052.59
2015	132.00	25.75	0.2874	43.32	1732.52
2016	137.00	25.92	0.2711	44.86	2179.36
2017	141.00	26.15	0.2829	43.73	2502.38
2018	140.00	27.58	0.2659	43.19	2296.71
2019	150.00	29.01	0.2643	43.28	1964.26
2020	164.00	26.17	0.2801	44.89	2269.63
2021	172.00	26.14	0.2759	44.42	2052.01
2022	190.00	28.24	0.2698	43.09	2361.96
2023	204.00	28.21	0.2786	42.06	2219.41
2024	199.00	30.60	0.2762	42.63	2459.51
2025	197.00	31.81	0.2805	43.12	2534.40

Chart: Correlation Matrix

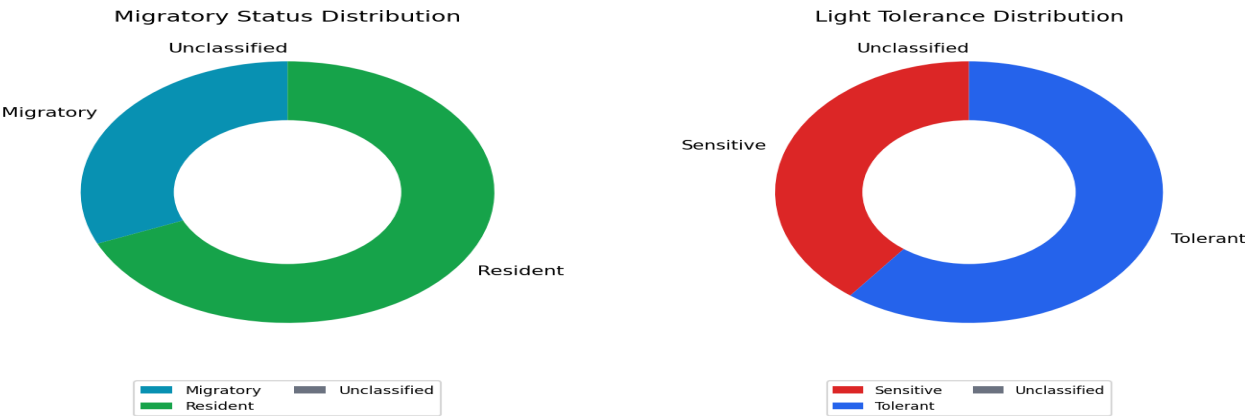


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Numerical Values: Correlation Coefficients

Metric	Pearson r
ALAN vs Richness	-0.0271
NDVI vs Richness	0.2895
LST vs Richness	0.1075
Precipitation vs Richness	0.1587

Chart: Migratory and Light Tolerance Distribution



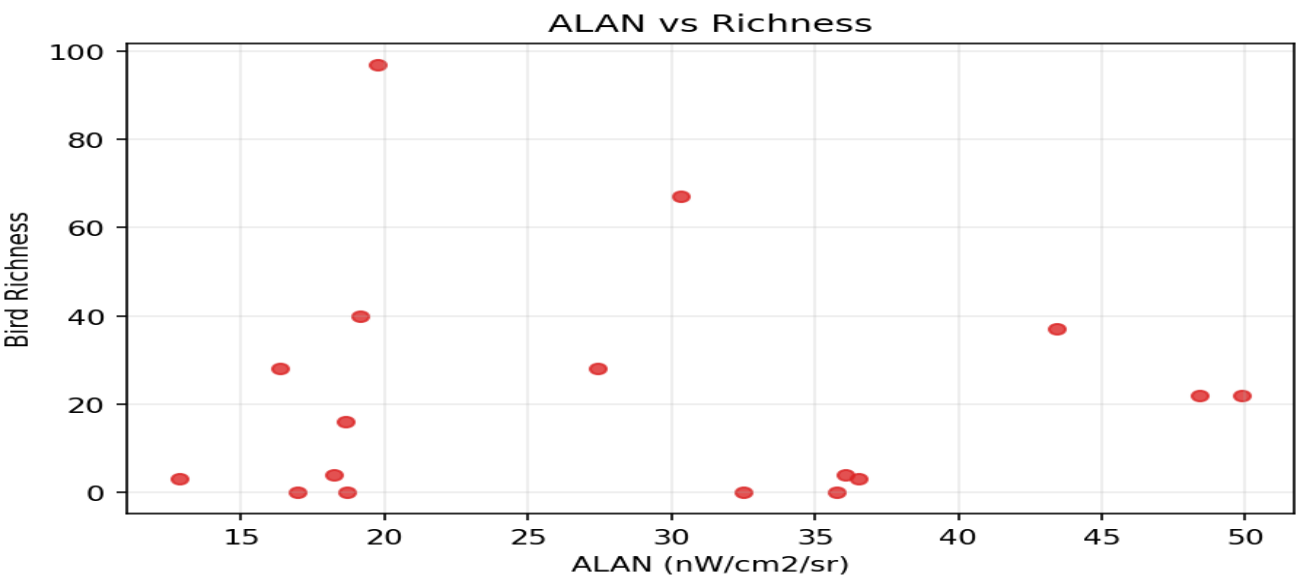
Numerical Values: Migration Distribution

Category	Count
Migratory	39
Resident	85
Unclassified	0

Numerical Values: Light Tolerance Distribution

Category	Count
Sensitive	49
Tolerant	75
Unclassified	0

Chart: ALAN vs Richness

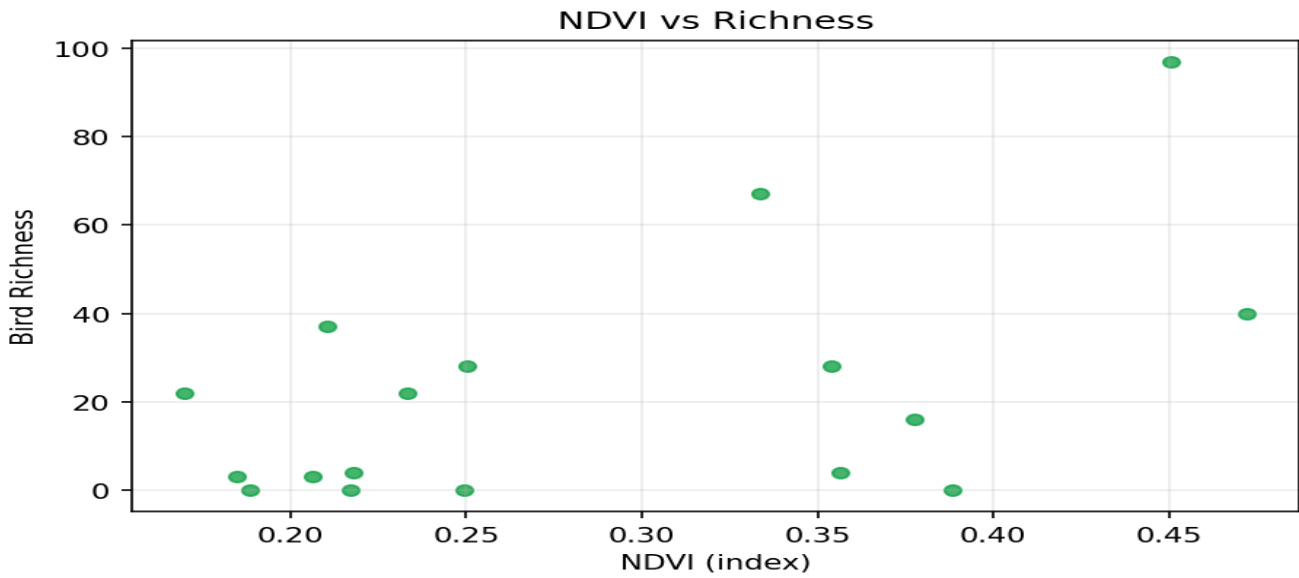


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Numerical Values: ALAN vs Richness

Site	Area	X	Richness
Caloocan	Caloocan	16.9669	0.00
Las Piñas	Las Piñas	18.6431	16.00
Makati	Makati	43.4287	37.00
Malabon	Malabon	18.6865	0.00
Mandaluyong	Mandaluyong	36.0768	4.00
Manila	Manila	48.4198	22.00
Marikina	Marikina	18.2037	4.00
Muntinlupa	Muntinlupa	19.1179	40.00
Navotas	Navotas	12.8473	3.00
Parañaque	Parañaque	30.3321	67.00
Pasay	Pasay	49.8610	22.00
Pasig	Pasig	35.7396	0.00
Pateros	Pateros	32.5225	0.00
Quezon City	Quezon City	19.7503	97.00
San Juan	San Juan	36.5149	3.00
Taguig	Taguig	27.4281	28.00
Valenzuela	Valenzuela	16.3417	28.00

Chart: NDVI vs Richness



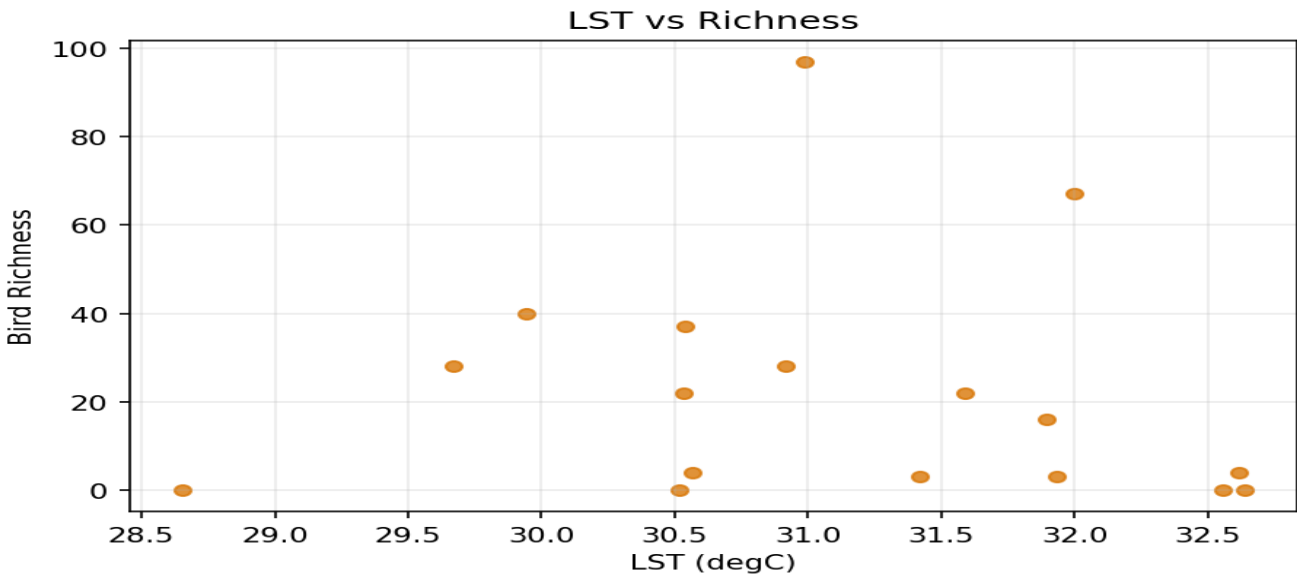
Numerical Values: NDVI vs Richness

Site	Area	X	Richness
Caloocan	Caloocan	0.3884	0.00
Las Piñas	Las Piñas	0.3776	16.00
Makati	Makati	0.2104	37.00
Malabon	Malabon	0.1886	0.00
Mandaluyong	Mandaluyong	0.2179	4.00
Manila	Manila	0.1699	22.00
Marikina	Marikina	0.3565	4.00
Muntinlupa	Muntinlupa	0.4719	40.00
Navotas	Navotas	0.1848	3.00

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Parañaque	Parañaque	0.3337	67.00
Pasay	Pasay	0.2332	22.00
Pasig	Pasig	0.2496	0.00
Pateros	Pateros	0.2173	0.00
Quezon City	Quezon City	0.4504	97.00
San Juan	San Juan	0.2063	3.00
Taguig	Taguig	0.3540	28.00
Valenzuela	Valenzuela	0.2501	28.00

Chart: LST vs Richness

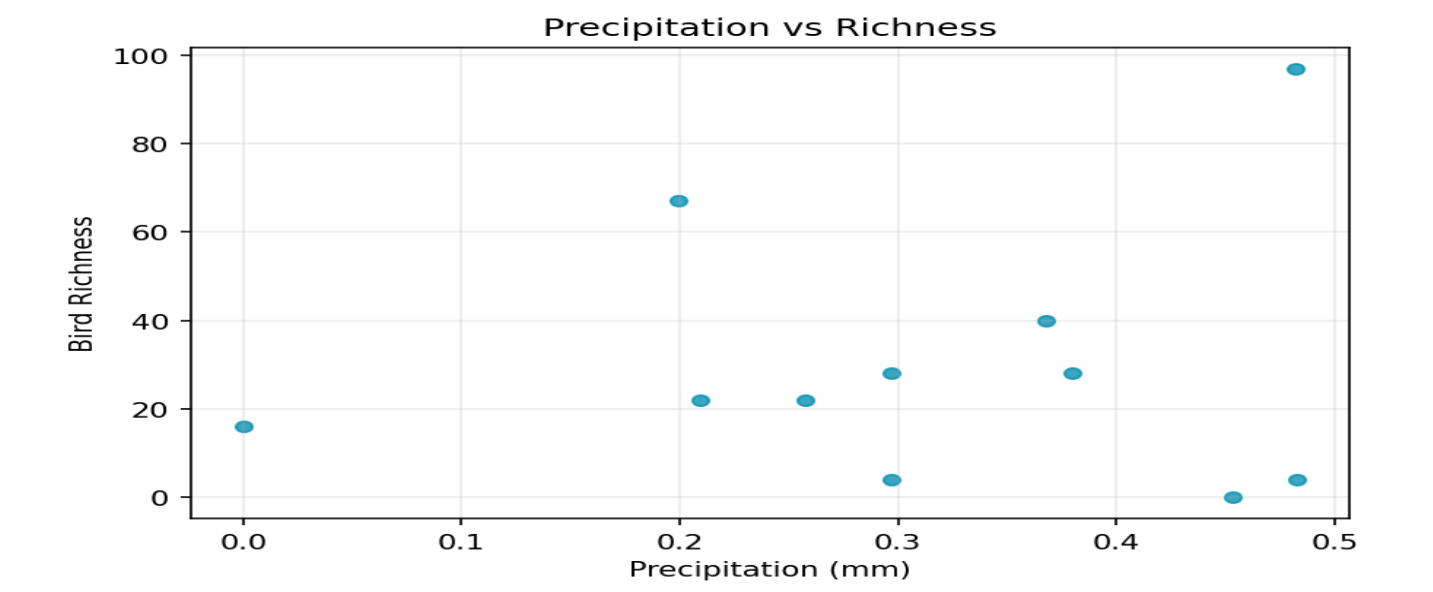


Numerical Values: LST vs Richness

Site	Area	X	Richness
Caloocan	Caloocan	30.5167	0.00
Las Piñas	Las Piñas	31.8972	16.00
Makati	Makati	30.5417	37.00
Malabon	Malabon	28.6532	0.00
Mandaluyong	Mandaluyong	32.6184	4.00
Manila	Manila	30.5324	22.00
Marikina	Marikina	30.5682	4.00
Muntinlupa	Muntinlupa	29.9428	40.00
Navotas	Navotas	31.4188	3.00
Parañaque	Parañaque	31.9967	67.00
Pasay	Pasay	31.5880	22.00
Pasig	Pasig	32.6377	0.00
Pateros	Pateros	32.5588	0.00
Quezon City	Quezon City	30.9879	97.00
San Juan	San Juan	31.9312	3.00
Taguig	Taguig	30.9138	28.00
Valenzuela	Valenzuela	29.6679	28.00

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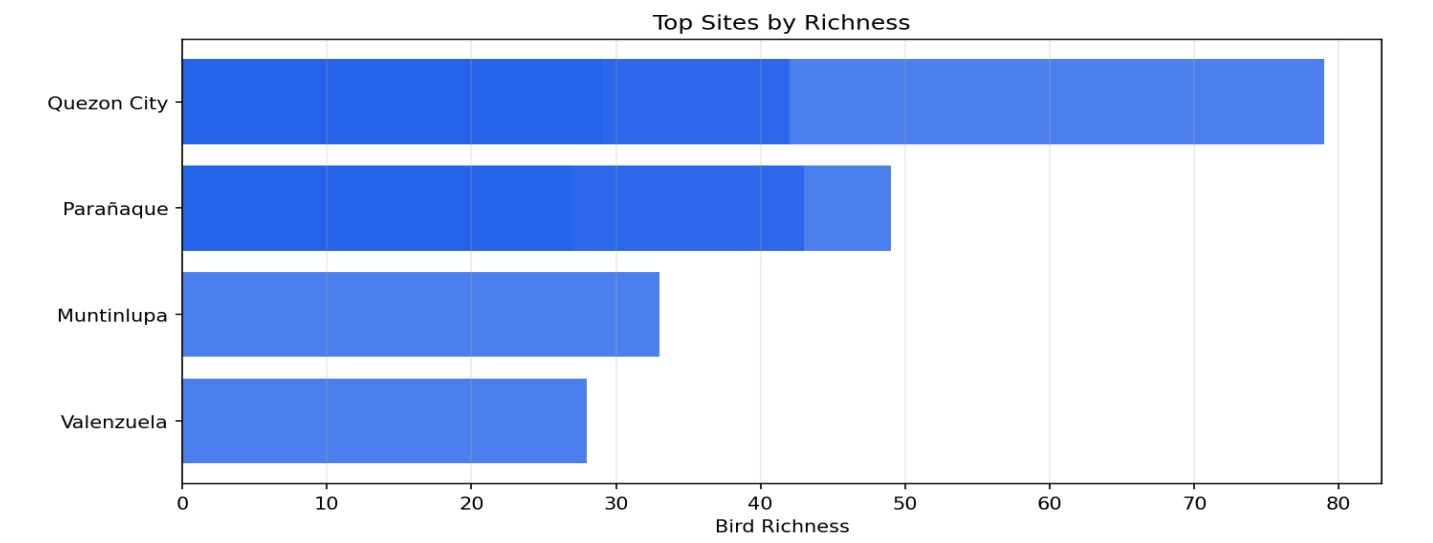
Chart: Precipitation vs Richness



Numerical Values: Precipitation vs Richness

Site	Area	X	Richness
Las Piñas	Las Piñas	0.0000	16.00
Mandaluyong	Mandaluyong	0.2972	4.00
Manila	Manila	0.2094	22.00
Marikina	Marikina	0.4827	4.00
Muntinlupa	Muntinlupa	0.3678	40.00
Parañaque	Parañaque	0.1991	67.00
Pasay	Pasay	0.2575	22.00
Pasig	Pasig	0.4534	0.00
Quezon City	Quezon City	0.4818	97.00
Taguig	Taguig	0.3800	28.00
Valenzuela	Valenzuela	0.2970	28.00

Chart: Top 10 Sites by Richness



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Numerical Values: Top Sites by Richness

Rank	Area	Richness
1	Quezon City	79
2	Parañaque	49
3	Parañaque	43
4	Quezon City	42
5	Muntinlupa	33
6	Quezon City	29
7	Quezon City	29
8	Valenzuela	28
9	Parañaque	27
10	Quezon City	26

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Model Diagnostics View

KPI Cards

Metric	Value
RMSE	0.0000
MAE	0.0000
R2	0.0000

Numerical Values: XGBoost Feature Importance (All Species)

Feature	Importance
No data	

Numerical Values: XGBoost Feature Importance (Light Sensitive)

Feature	Importance
No data	

Numerical Values: XGBoost Feature Importance (Light Tolerant)

Feature	Importance
No data	

Numerical Values: XGBoost Feature Importance (Migratory)

Feature	Importance
No data	

Numerical Values: XGBoost Feature Importance (Resident)

Feature	Importance
No data	

Numerical Values: ConvLSTM Actual vs Predicted (All Species)

Year	Actual	Predicted
No data		

Numerical Values: ConvLSTM Actual vs Predicted (Light Sensitive)

Year	Actual	Predicted
No data		

Numerical Values: ConvLSTM Actual vs Predicted (Light Tolerant)

Year	Actual	Predicted
No data		

Numerical Values: ConvLSTM Actual vs Predicted (Migratory)

Year	Actual	Predicted
No data		

Numerical Values: ConvLSTM Actual vs Predicted (Resident)

Year	Actual	Predicted
No data		

Systems Recommendations: KBA/PA Audit (Core)

Site	Type	Species	Sens.Cnt	Sens.%	Light	Eff.	Status
Manila Bay Beach Resort	PA	12	1	8.3	63.4	22.7	Critical

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Luneta National Park	PA	29	6	20.7	78.9	24.2	Critical
Ninoy Aquino Parks and Wildlife Center	PA	41	10	24.4	48.2	35.8	Critical
Las Piñas-Parañaque Wetland Park	PA	64	21	32.8	17.5	52.6	At Risk
Manila Bay	KBA	92	35	38.0	11.6	56.0	At Risk

Systems Recommendations: KBA/PA Audit (Environment)

Site	Mean NDVI	Max LST	Precip	Cells	Snapshot
Manila Bay Beach Resort	0.1952	36.69	209.70	45	2025-12
Luneta National Park	0.2073	37.42	27.01	8	2025-12
Ninoy Aquino Parks and Wildlife Center	0.3363	37.42	39.98	9	2025-12
Las Piñas-Parañaque Wetland Park	0.2970	36.23	198.71	30	2025-12
Manila Bay	0.2370	37.42	4151.56	874	2025-12